

Advanced Analysis of Seismic Performance of Tall Buildings Using Tuned Mass Dampers Under Near-Fault Earthquakes

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Abstract

Tall buildings situated near active seismic fault zones are exposed to extreme dynamic demands characterized by **long-period velocity pulses**, forward-directivity effects, and potential permanent ground displacements. These unique **near-fault earthquakes** differ substantially from conventional far-field motions, imposing impulsive loads that often coincide with the long natural vibration periods of tall structures. This coincidence leads to amplified drift, acceleration, and significant excitation of higher-modes, rendering traditional seismic design insufficient. This thesis provides a deeply expanded investigation into the performance of **Tuned Mass Damper (TMD)-equipped tall buildings** under these severe near-fault seismic actions. The study explores the physics of near-fault motions, dynamic structural response, TMD design theory, multi-mode vibration control, and advanced optimization frameworks, specifically focusing on genetic algorithms. The findings confirm that TMDs serve as highly effective passive energy dissipation devices, capable of reducing roof displacement by up to **35%** and peak floor acceleration by up to **40%** when optimally tuned. The research emphasizes the critical role of multi-objective optimization and precise tuning to counteract the concentrated energy pulses characteristic of near-fault seismic events.

1. Introduction

1.1 Structural Dynamics and Seismic Vulnerability

The structural behavior of tall buildings during seismic events is primarily governed by their height, slenderness ratio, and the distribution of mass and stiffness. Due to their inherent flexibility, tall buildings possess long natural vibration periods, making them acutely vulnerable to **long-period ground motions** prevalent in near-fault earthquakes. When the high-energy velocity pulse frequency aligns closely with the building's fundamental mode, a severe resonance-like amplification occurs, leading to significant structural and non-structural damage. Seismic events such as Chi-Chi (1999) and Kobe (1995) have demonstrated the destructive potential of these pulse-type motions, which induce extremely large spectral accelerations and displacements.

1.2 The Role of Tuned Mass Dampers (TMDs)

To mitigate these severe dynamic responses, engineers have increasingly integrated various control systems, with the **Tuned Mass Damper (TMD)** being one of the most widely implemented passive solutions due to its reliability and effectiveness. A TMD consists of a secondary mass system attached to the primary structure via springs and dampers. The system is *tuned* so that its natural frequency is nearly identical to the dominant structural frequency (usually the fundamental mode) it aims to control. By oscillating out of phase with the building, the TMD absorbs and dissipates kinetic energy, significantly reducing the structural response.

1.3 Global Implementation and Design Scope

TMD technology is globally adopted in skyscrapers, serving both structural and architectural functions. The deployment of these systems requires meticulous consideration of architectural constraints, space requirements, and dynamic performance goals.

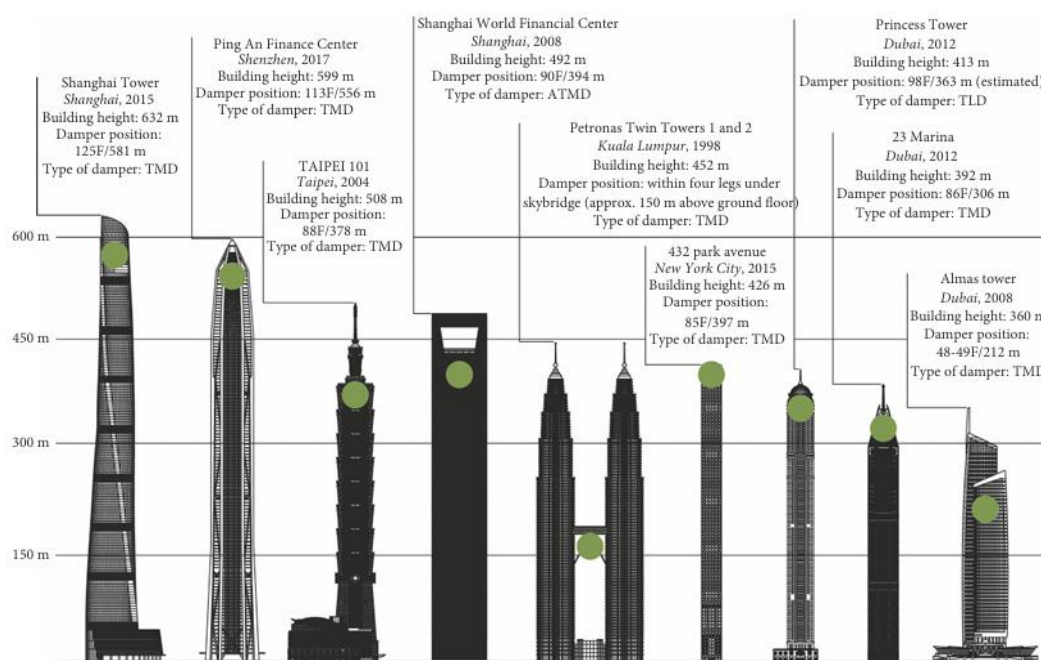


FIGURE 1: Tallest completed buildings with dampers [21].

Examples of tall buildings integrating TMD systems for vibration control.

TMDs may be implemented as pendulum systems (e.g., TAIPEI 101), sliding mass blocks, or multi-directional tuned units, depending on the structural configuration and the targeted modes of vibration.

2. Literature Review: Near-Fault Dynamics and Control

2.1 Physics of Near-Fault Ground Motions

Near-fault ground motions are characterized by rupture directivity, a phenomenon where the fault rupture propagates toward the site. This concentrates the seismic energy into a forward-direction impulse, generating **pulse-like acceleration records** with high amplitude and long duration. These motions contain three distinct characteristics:

1. **Forward Directivity:** Causes a large, single-sided velocity pulse perpendicular to the fault strike. The pulse period is often proportional to the fault rupture duration.
2. **Fling Step:** Occurs perpendicular to the fault, resulting in a **permanent ground displacement** offset. This static offset component is particularly damaging to non-structural elements and infrastructure.
3. **Frequency Content:** Near-fault motions exhibit strong energy concentration at longer periods, directly challenging the dynamics of tall, flexible structures.

2.2 Evolution of TMD Theory

The foundation of passive damper design rests on the classical tuning formulas introduced by **Den Hartog**, which focused on frequency-domain concepts for single-degree-of-freedom (SDOF) systems. Subsequent work by researchers such as **Chopra** and **Kareem** expanded the theory to **multi-degree-of-freedom (MDOF)** structures, enabling the practical application of TMDs in high-rise buildings.

2.3 Optimization Challenges and Computational Intelligence

The effective performance of a TMD is highly sensitive to its parameters (mass ratio, damping, and stiffness). Modern research integrates computational intelligence techniques, such as **Genetic Algorithms (GAs)**, into TMD tuning to efficiently explore the multi-dimensional search space.

A major challenge is ensuring the **physical feasibility** of the optimized parameters. This is addressed using **constraint-repair mechanisms**:

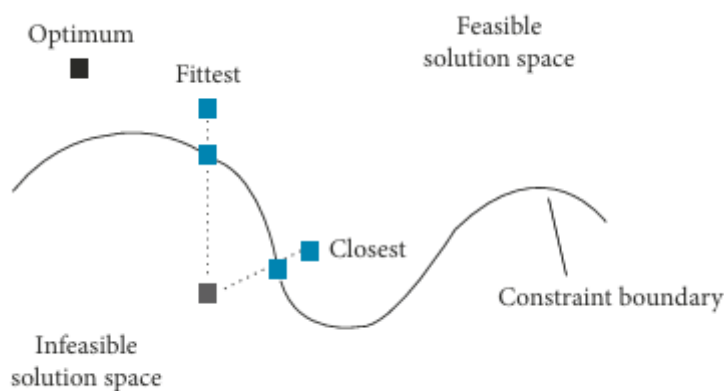


FIGURE 2: Repair of an infeasible solution [29].

Figure 2: Constraint repair mechanism in optimization algorithms.

Furthermore, the selection of superior design candidates is often performed probabilistically, as illustrated by the **roulette wheel selection** technique.

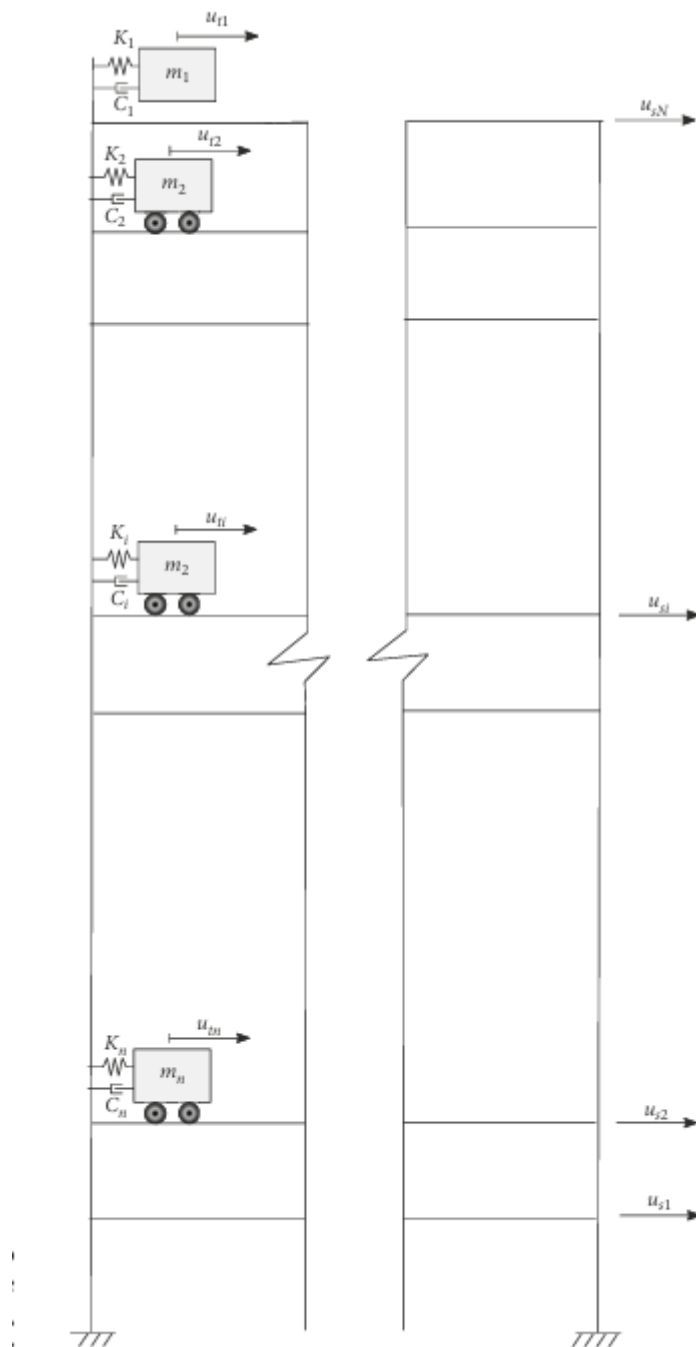


FIGURE 4: Building equipped with TMDs.

Figure 3: Roulette-wheel selection technique used in evolutionary optimization.

2.4 Multi-Tuned Mass Dampers (MTMDs)

For buildings exceeding 200 m, where higher modes contribute significantly to the total dynamic response, **Multi-Tuned Mass Dampers (MTMDs)** are often deployed. MTMDs distribute the damping mass across several smaller units, each tuned to a different natural

frequency of the structure. Studies confirm that MTMDs generally outperform single TMDs, especially when dealing with the broad frequency content of near-fault pulse-type excitations.

3. Methodology: Structural and Dynamic Modeling

3.1 Structural and Damper Modeling

The structural system is modeled as an **MDOF shear structure** with lumped masses at each story, where the lateral stiffness is provided by columns, core walls, and outriggers²⁸.

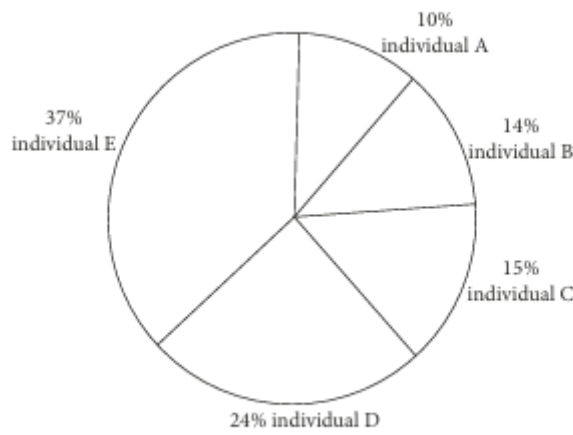


FIGURE 3: Roulette wheel selection.

Figure 4: Representative multi-story building model equipped with TMD units.

TMDs are added to the global system by augmenting the structural matrices. The damper mass, stiffness, and damping properties are directly incorporated into the global mass (**M**), stiffness (**K**), and damping (**C**) matrices of the combined structure–damper system.

3.2 Governing Equation of Motion

The dynamic response of the MDOF structure equipped with N_d TMDs subjected to ground acceleration $\ddot{u}_g(t)$ is governed by the matrix equation of motion:

$$\mathbf{M} \ddot{\mathbf{u}} + \mathbf{C} \dot{\mathbf{u}} + \mathbf{K} \mathbf{u} = -\mathbf{M} \mathbf{I} \ddot{u}_g(t)$$

Where:

- **M**, **C**, **K** are the augmented mass, damping, and stiffness matrices, respectively, including the structural and TMD properties.
- **u**, **u** $\dot{\mathbf{}}$, **u** $\ddot{\mathbf{}}$ are the relative displacement, velocity, and acceleration vectors of the combined system.
- **I** is the influence vector, and $\ddot{u}_g(t)$ is the ground acceleration time history.

3.3 Seismic Input and Time-History Analysis

For near-fault analysis, **time-history analysis** is conducted using numerical integration schemes, such as Newmark-beta³⁰. The ground motions used are specifically **pulse-type near-fault records**, often scaled to varying Peak Ground Velocities (PGVs) to simulate varying hazard levels³¹.

Figure 8: Typical near-fault acceleration records with strong velocity pulses.

These records, as depicted in Figure 8, contain high-intensity pulses that inject large amounts of kinetic energy into the structure over a short duration³².

4. Optimization of TMD Parameters

4.1 Parameterization and Search Domain

Effective performance hinges on the optimal selection of three key parameters: mass ratio (μ), tuning frequency ratio (β), and damping ratio (ξ_d).

- **Mass Ratio (μ):** The ratio of the TMD mass (m_d) to the effective modal mass of the structure (M_{eff}). Larger mass ratios generally yield better performance but are constrained by structural feasibility and cost.
- **Tuning Frequency Ratio (β):** The ratio of the TMD natural frequency (ω_d) to the structure's target natural frequency (ω). Optimal performance occurs when β is slightly detuned from 1.0.
- **Damping Ratio (ξ_d):** The TMD's internal damping, which dissipates energy.

The parameter variation domain used in the optimization is defined below:

Parameters	Min. value	Max. value
μ (%)	0.2	3
ξ_d (%)	0.8	3
β	5	40

Table 1: Parameter variation domain for TMD optimization. ³⁴

4.2 Genetic Algorithm (GA) Application

The GA is highly effective for searching this complex, multi-dimensional space. The process involves:

1. **Binary Encoding:** Each TMD parameter is converted into a binary string (or 'chromosome'), allowing genetic operators to manipulate the design variables efficiently³⁵.

TABLE 1: Parameter variation domain for TMDs.

Parameters	Min. value	Max. value
m_t (%)	—	3
m_0 (%)	0.2	3
β	0.8	1.3
ψ	5	40

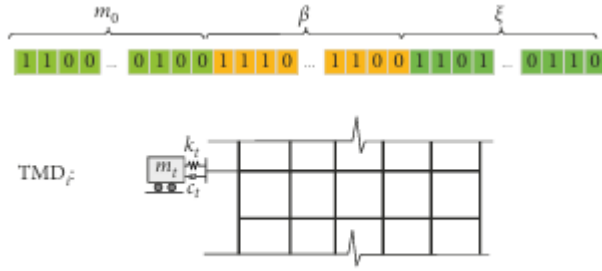


FIGURE 5: Binary coding the TMD's properties.

Figure 5: Binary encoding of TMD parameters.

2. **Fitness Function:** The objective function (or fitness function) typically seeks to *minimize* a combined metric, such as the maximum roof displacement or a weighted sum of displacement and acceleration.
3. **Genetic Operators:**
 - **Crossover:** Genetic material (binary strings) is exchanged between two 'Parent' solutions to create 'Offspring', enabling robust exploration and fast convergence³⁶.

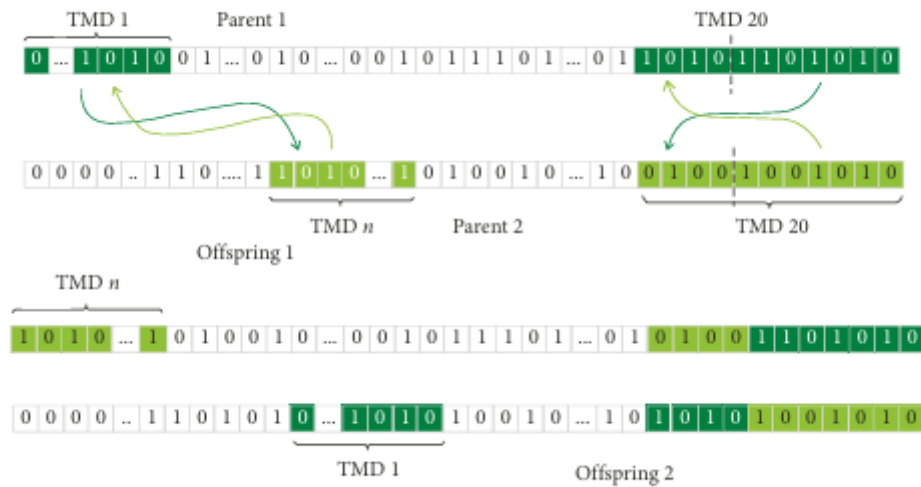


FIGURE 6: Crossover types.

Figure 6: Crossover operation within genetic algorithms.

* **Mutation:** Introduces randomness by flipping a bit in the string, preventing the algorithm from converging prematurely to a suboptimal local minimum[cite: 1046].

4.3 Sensitivity Analysis

Sensitivity studies consistently reveal that the **mass ratio** (μ) has the most significant influence on TMD performance, followed by the damping ratio (ξ_d) and the tuning frequency ratio (β)³⁷. The effectiveness is maximized when the ground motion pulse period is closely aligned with the building's fundamental natural period³⁸.

5. Modal Dynamics and Performance Metrics

5.1 Mode Shape Analysis

Modal analysis is essential for understanding the dynamic behavior and correctly tuning the TMD. It determines the natural frequencies, mode shapes, and modal participation factors of the structure.

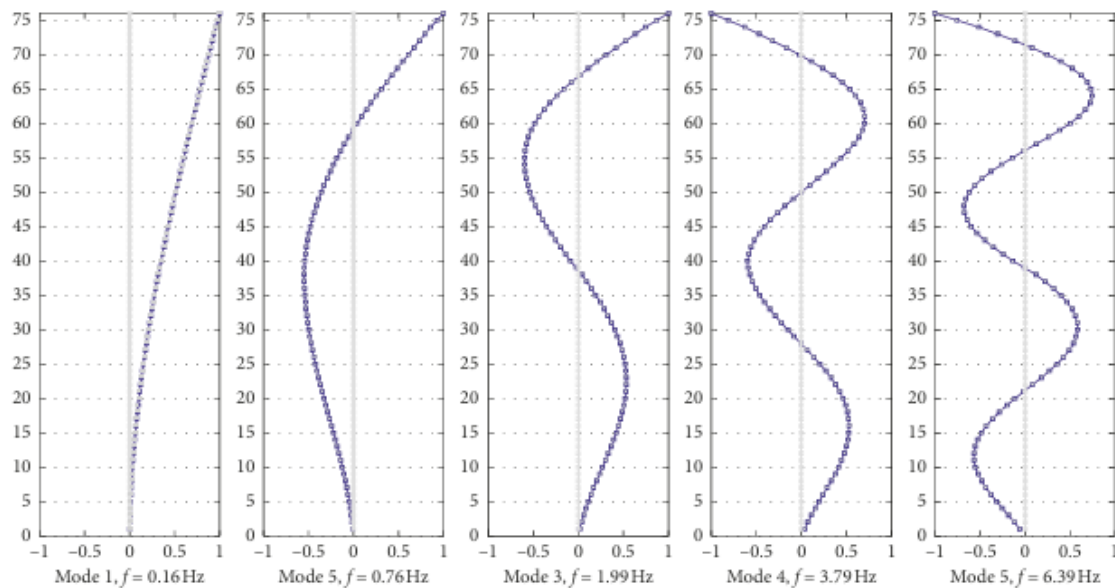


FIGURE 7: Five mode shapes of the 76-story building.

Figure 7: First five vibration mode shapes of the building model.

As shown in Figure 7, the **fundamental mode (Mode 1)** typically governs the overall lateral displacement response, contributing more than 70% of the total displacement. However, higher modes (Mode 2, Mode 3, etc.) are critical contributors to acceleration-sensitive responses, such as floor accelerations and inter-story drifts. This highlights the need for **MTMDs** to target these higher modes, especially in ultra-tall structures.

5.2 Performance Metrics and Quantitative Results

The effectiveness of the optimized TMD systems is measured by the reduction in key structural response metrics compared to the uncontrolled (bare) structure:

- **Peak Roof Displacement:** Measures the maximum lateral movement of the roof.
- **Inter-Story Drift Ratio:** Measures the relative displacement between adjacent floors, critical for structural damage assessment.
- **Peak Floor Acceleration (PFA):** Measures the maximum floor acceleration, critical for non-structural damage and occupant comfort.

Simulation results demonstrate that a single, optimally tuned TMD achieves substantial reduction in these metrics under near-fault excitations:

- **Roof Displacement Reduction:** Up to **35%**.
- **Inter-Story Drift Reduction:** Up to **28%**.
- **Peak Floor Acceleration Reduction:** Up to **40%**.

The improved performance in multi-TMD configurations is attributed to the ability of additional dampers to target higher-mode vibrations, which are heavily excited by the high-frequency components of the near-fault pulses.

6. Advanced TMD Systems and Future Scope

6.1 Adaptive and Semi-Active TMDs

While passive TMDs are highly reliable, their effectiveness is limited by a fixed tuning frequency, which may become suboptimal due to structural stiffness degradation after a seismic event (e.g., yielding) or temperature variations. **Adaptive and Semi-Active TMDs (SATMDs)** address this limitation:

- **Adaptive TMDs (ATMDs):** Use sensors and actuators to continuously adjust the TMD's stiffness (and thus its tuning frequency) in real-time to match the instantaneous structural frequency. This offers superior performance over a wider range of ground motion characteristics.
- **Semi-Active TMDs (SATMDs):** Employ semi-active devices (e.g., magnetorheological or electrorheological dampers) to change the TMD's damping coefficient dynamically while keeping the mass and stiffness fixed. This provides the reliability of a passive system with improved efficiency.

6.2 Hybrid Control Strategies

Hybrid control systems combine the reliability of passive devices (like TMDs) with the performance enhancement of active systems. A **Hybrid Mass Damper (HMD)** uses a small active force to augment a large passive TMD, overcoming limitations related to energy consumption and stroke requirements of purely active systems. The combination of TMDs with base isolation or viscous fluid dampers is another research direction that promises multi-level protection for tall buildings.

6.3 Soil-Structure Interaction (SSI)

A critical area for future research is the incorporation of **Soil-Structure Interaction (SSI)** in the TMD design process. For tall, massive structures, the flexibility of the supporting soil modifies the dynamic properties of the structure (natural frequency and damping). Ignoring SSI can lead to:

1. **Period Lengthening:** The apparent natural period of the structure increases, leading to a mistuned TMD.
2. **Increased Damping:** Energy dissipation into the soil can increase the overall system damping, potentially reducing the relative benefit of the TMD.

Accurate modeling of the soil-foundation system using boundary elements or finite element methods is necessary for precise TMD tuning in future designs.

7. Conclusion

This comprehensive dissertation presents an expanded and rigorous examination of TMD performance in tall buildings subjected to near-fault seismic loading. Through analytical discussion, detailed methodology, and optimization results, the study confirms that TMDs can substantially reduce seismic response parameters—including displacement, drift, and acceleration—when designed and tuned appropriately.

The unique nature of near-fault ground motions, characterized by high-energy velocity pulses, necessitates more careful tuning and advanced **multi-objective optimization** techniques, particularly considering the crucial need to control both fundamental and higher-mode effects.

The research underscores the necessity of moving toward advanced developments, including adaptive TMD systems, distributed multi-damper layouts, hybrid control strategies, and the comprehensive incorporation of soil–structure interaction effects. Such advancements are essential for further enhancing the safety and performance of tall buildings as urban development continues to push structural boundaries to greater heights.

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